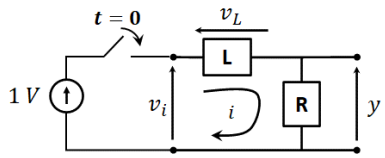
EE414 assignment 3

Due 01/05/2026
Submit one **pdf** file on Moodle
Late submissions lose 10% per day
Submissions not accepted after 04/05/2026
Marks: 10% of module
This assignment should be typed, not handwritten. Your parameters are on the last page.

Your individual parameters are on page 3.

In this assignment, **time is in picoseconds**, not seconds.



(Q1) Use your electric circuit skills to show that the ODE model for the voltage across the resistor, y, is:

$$\frac{dy}{dt} = \frac{1}{\tau}(1 - y) \text{ where } \tau = \frac{L}{R}$$

(Q2) Use your mathematical skills to find the analytic solution to the IVP:

$$\frac{dy}{dt} = \frac{1}{\tau}(1 - y),\ y(0) = 0$$

Hint. Separation of variables or Laplace transforms or Solution by inspection.

(Q3) Code Euler's method (see slide 10 of notes 5) and solve this IVP using Euler's method to estimate $y(40), y(80), \ldots y(240)$ using an integration step size:

- $h = 40$ ps
- $h = 20$ ps
- $h = 10$ ps

Tabulate your results as shown below. Use your Q2 solution to find $y(t_i)$. Use two significant digits for t_i and E_t, use five significant digits for $\hat{y}(t_i)$ and $y(t_i)$. Format E_t in scientific notation. Ensure your table is on one page. You may want to use landscape format for that page. Include a copy of your code.

$h = 40$ ps					$h = 20$ ps					$h = 10$ ps				
i	t_i	$\hat{y}(t_i)$	$y(t_i)$	E_t	i	t_i	$\hat{y}(t_i)$	$y(t_i)$	E_t	i	t_i	$\hat{y}(t_i)$	$y(t_i)$	E_t
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
										1	10	?	?	?
					1	20	?	?	?	2	20	?	?	?
										3	30	?	?	?
1	40	?	?	?	2	40	?	?	?	4	40	?	?	?
										5	50	?	?	?
					3	60	?	?	?	6	60	?	?	?
										7	70	?	?	?
2	80	?	?	?	4	80	?	?	?	8	80	?	?	?
:	:	:	:	:	:	:	:	:	:	:	:	:	:	:

(Q4) Repeat Q3 using RK2. Include a copy of your code.

(Q5) Repeat Q3 using RK4. Include a copy of your code.

(Q6) Repeat Q3 using Simulink. Include a copy of your Simulink model.

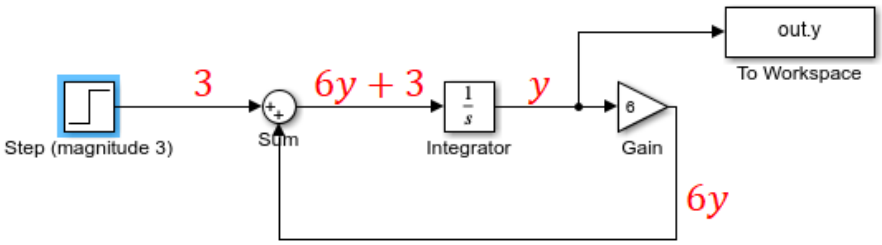
(Q7) Repeat Q4 using Simulink.

(Q8) Repeat Q5 using Simulink.

(Q9) Discuss your Q3 to Q8 results. How does RK1, RK2 and RK4 compare in terms of accuracy and computation? How do your Simulink results compare to your coded results? Your discussion should be specific to **your** results i.e. your E_t

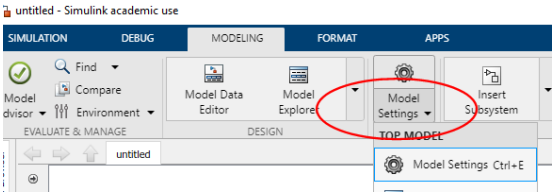
Grading: Q1, …, 8 (10% each), Q9 (20%)
Include your code and a screenshot of your Simulink model.

<u>Simulink</u>
Your Simulink model will need an *integrator* block, a *gain* block, a *sum* block, a *step* block and a *toWorkspace* block. Simulink represents an ODE graphically. For example, $\frac{dy}{dt} = 6y + 3$ is represented as:

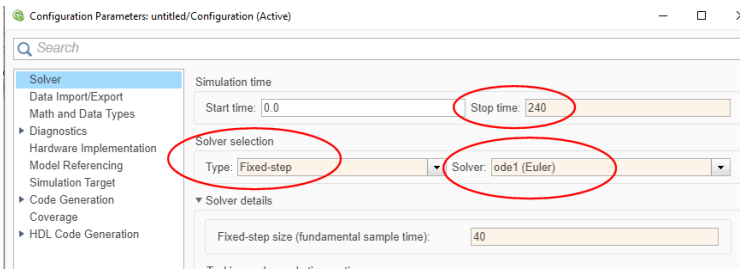


The *toWorkspace* block saves y to the MATLAB workspace. You need to set its *Sample Time*.

<u>How to select RK1, RK2 or RK4 in Simulink</u>
From the MODELLING tab, select Model Settings



Set the Stop Time to 240
Set the Solver Selection Type to Fixed-Step
Set the Solver to ODE1 or ODE2 or ODE4



Use of AI is prohibited for EE414 CA

ID	R (MΩ)	L (μH)
20353843	1	110
22439376	2	240
18478834	3	270
21455956	4	580
25256066	5	530
25256067	6	600
20363191	7	900
21349661	8	570
20405256	9	900
22437852	1	130
25256065	2	250
25256063	3	450